

NITESH KUMAR LODH

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LeetCode: NiteshKumarLodh

PROFILE SUMMARY

Computer Science undergraduate at PDEU (graduating 2027) with demonstrated experience building and shipping full-stack, real-time, and AI-powered applications across web and mobile platforms. Projects span healthcare operations, fintech, computer vision, and cybersecurity — reflecting both technical breadth and hands-on depth. Security-conscious with practical knowledge of OWASP Top 10, responsible disclosure experience, and exposure to offensive security tooling.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES6+), Java, C, Dart, Bash / Shell scripting
- **Backend:** Node.js, Express, FastAPI, WebSockets (Socket.IO), REST API design, JWT, OOP, MVC architecture
- **Frontend:** React (Vite + Tailwind), HTML5, CSS3, Flutter / Dart
- **Databases:** MongoDB, PostgreSQL / Supabase, MySQL / MariaDB — schema design, normalisation, CRUD, transactions
- **AI / ML:** TensorFlow Lite, face-api.js, prompt engineering, transfer learning (EfficientNetB3), OpenCV
- **Security:** Kali Linux, Burp Suite, Nmap, Wireshark, SQLMap, Hydra, Metasploit, OWASP Top 10, 802.11 / DeAuth
- **Tools:** Git, GitHub, Redis, Render, Vercel, Razorpay, ImageKit

EDUCATION

B.Tech — Computer Science Engineering · Pandit Deendayal Energy University (PDEU) 2023 – 2027

- CGPA: 7.37 | Coursework: Data Structures & Algorithms · DBMS · Operating Systems · Computer Networks · AI

XII (NEB) · Manimukunda HSS / College 2023

- Aggregate: 74.25%

PROJECTS

AarogyaLink *React · Node.js · Express · MongoDB · Socket.IO · JWT* Apr 2025

- Full-stack healthcare platform for doctor appointment booking and real-time token queue management, designed for clinical operations in Nepal.
- Live queue updates via Socket.IO WebSockets; JWT-based role authentication for patients, doctors, and administrators.
- RESTful API with input validation and RBAC; schema structured for concurrent multi-clinic deployments.

GoRent *React · Node.js · Express · MongoDB · Leaflet.js / OSM · ImageKit · JWT* Apr 2025

- Car rental booking platform with interactive map-based location selection (OpenStreetMap / Leaflet.js), vehicle browsing, and post-ride feedback.
- ImageKit CDN integration for optimised vehicle image delivery; full JWT auth with protected booking workflows and admin fleet dashboard.

Splitora — Breach 2026 FinTech Hackathon *FastAPI · PostgreSQL / Supabase · React / Vite / Tailwind · Razorpay*
Mar 2025

- Smart expense splitting and settlement system built with team Logic Lords (3 members); backend architected with FastAPI and PostgreSQL on Supabase, deployed on Render / Vercel.
- Designed a graph-theory based Combined Settlement Algorithm that minimises the number of transactions needed to settle group debts.
- Razorpay payment gateway integration and a Google Pay PDF transaction importer for automatic expense parsing.

NKLStream *Node.js · Socket.IO · WebRTC · JavaScript* Mar 2025

- Synchronised YouTube co-watching platform with real-time chat and video call support; sub-100 ms playback latency via Socket.IO.
- RESTful session management with input validation and auth handling; deployed to Render with full route documentation.
- Companion Android app (NKLStreamYT-App) published separately with a mobile-optimised UI.

Real-Time Facial Analytics System *React · face-api.js · Node.js · Express · MongoDB* Mar 2025

- Full-stack system for real-time facial detection, recognition, and analytics via live webcam stream using face-api.js on the React frontend.
- Node.js / Express backend with MongoDB for storing detection logs and analytics data; REST API for frontend-backend communication.

Food Scanner *Flutter / Dart · TensorFlow Lite · EfficientNetB3* Mar 2025

- AI-powered mobile app for real-time food image classification using a TFLite INT8 quantised model trained with EfficientNetB3 transfer learning.
- Health scoring and nutritional analysis with a curated Indian / Nepali food dataset; optimised for fully on-device inference.

NKL Launcher *Flutter / Dart · Android Platform Channels* Apr 2025

- Minimalist Android launcher with dual calendar support (AD / BS), built with Flutter and platform channels for system-level Android integration.
- Clean, distraction-free UX with custom widget and app-drawer management; inspired by Minimalist Phone.

DeAuth Tool *Shell / Bash · Kali Linux · IEEE 802.11* Apr 2025

- Shell-based Wi-Fi deauthentication tool for educational penetration testing; demonstrates 802.11 frame injection and wireless attack methodology.
- Built for controlled lab environments; illustrates understanding of wireless protocol vulnerabilities and ethical offensive security tooling.

Sabji Mandi Nandanagar *HTML5 · CSS3 · JavaScript · SEO* May 2025

- SEO-optimised, mobile-first responsive website for a fresh vegetable market in Nepal; features interactive gallery, customer reviews, and location integration.
- Structured data markup for local business discoverability; 2 GitHub stars, live deployment.

Also: Hospital Management System (Java · MySQL · JDBC · RBAC) · Health Coach AI Agent (Python · AI API, recognised by Ashna AI) · Photo Editing CLI (Python · U2Net AI background removal)

EXPERIENCE & LEADERSHIP

Tech & Social Media Head · EmpowerU, PDEU 2025

- Managed technical infrastructure for the organisation; led Energy Auction event with 50+ student participants.
- Researched and adopted tooling to improve digital workflows and social media outreach.

Teacher & Mentor · Shree Janajagriti Ma. Vi. 2024

- Delivered structured lessons on academic subjects and technology fundamentals, improving secondary-level cohort performance.

Technical Head · Manimukunda HSS / College 2021 – 2023

- Led AV and technical operations for the Inter-College Quiz Competition and annual Teej Program (200+ attendees).

ACHIEVEMENTS & CERTIFICATIONS

- **Responsible Disclosure:** Discovered and reported a database exposure vulnerability in the PDEU official website to university IT administration.
- **Ashna AI Recognition:** Health Coach AI Agent integrated into Ashna AI's live production platform and publicly showcased.
- **Breach 2026 FinTech Hackathon:** Built and presented Splitora (smart group expense settlement) as backend lead of team Logic Lords.
- **IBM Python 101 for Data Science** — certified by IBM / Cognitive Class (2026).
- **Open Source & Portfolio:** 21 public repositories on GitHub; personal portfolio live at nitesh-kumar.com.np.